



The Network, Internet and Web

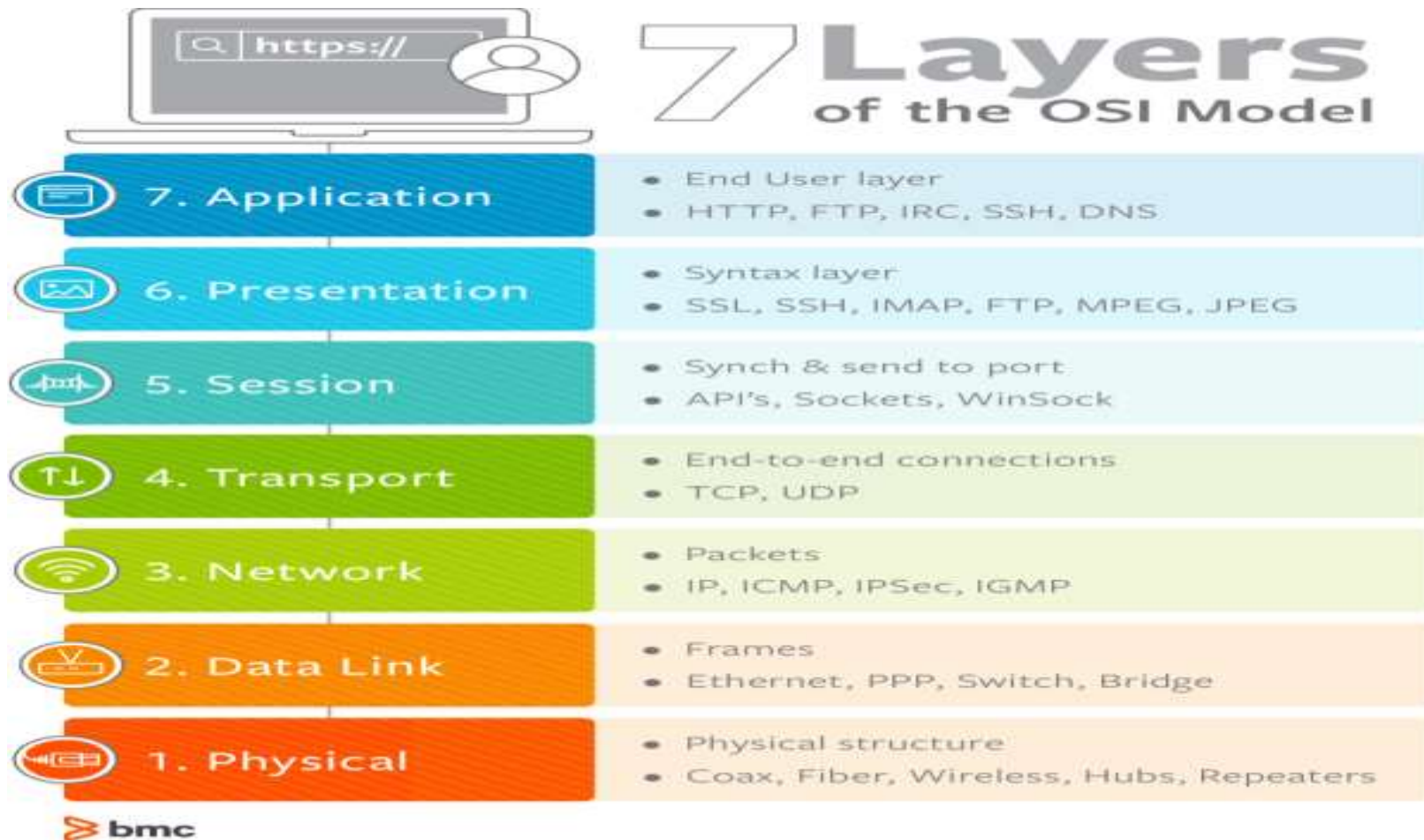


The Network: Technology Background

■ Network

- ❖ A network is formed when 2 or more computers are linked together.
- ❖ Every computer or device on a network can send and receive data from any of the other computers or devices connected to the network.
- ❖ A computer which is not part of a network is called a stand-alone computer.

The Network: OSI Model





The Internet: Technology Background

■ Internet

- ❖ Interconnected network of thousands of networks and millions of computers
- ❖ Links businesses, educational institutions, government agencies, and individuals

■ World Wide Web (Web)

- ❖ One of the Internet's most popular services
- ❖ Provides access to billions, possibly trillions, of Web pages



Evolution of the Internet 1961–Present

■ Innovation Phase, 1964–1974

- ❖ Creation of fundamental building blocks
 - Packet switching, TCP/IP, Client-server computing

■ Institutionalization Phase, 1975–1995

- ❖ Large institutions provide funding and legitimization creating ARPANET

■ Commercialization Phase, 1995–present

- ❖ Private corporations take over, expand Internet backbone and local service



The Internet: Key Technology Concepts

■ Internet defined as network that:

- ❖ Uses IP addressing
- ❖ Supports TCP/IP
- ❖ Provides services to users, in manner similar to telephone system

■ Three important concepts:

- ❖ Packet switching
- ❖ TCP/IP communications protocol
- ❖ Client/server computing



Packet Switching

- Slices digital messages into packets
- Sends packets along different communication paths as they become available
- Reassembles packets once they arrive at destination
- Uses routers
 - ❖ Special purpose computers that interconnect the computer networks that make up the Internet and route packets
 - ❖ **Routing algorithms** ensure packets take the best available path toward their destination
- **Less expensive, wasteful than circuit-switching**



Packet Switching

I want to communicate with you.

Original text message

0010110110001001101110001101

Text message digitized into bits

01100010 10101100 11000011

Digital bits broken into packets

0011001 10101100 11000011

Header information added to each packet indicating destination, and other control information, such as how many bits are in the total message and how many packets

Figure 3.3, Page 117



TCP/IP

■ Transmission Control Protocol (TCP)

- ❖ Establishes connections among sending and receiving Web computers
- ❖ Handles assembly of packets at point of transmission, and reassembly at receiving end

■ Internet Protocol (IP)

- ❖ Provides the Internet's addressing scheme to deliver packets

■ Four TCP/IP architecture layers

- ❖ Network interface or link layer responsible for sending and receiving packets
- ❖ Internet layer responsible for addressing and routing
- ❖ Transport layer handles host-to-host communication
- ❖ Application layer provides an array of applications to users such as HTTP, FTP, SMTP.

The TCP/IP Architecture and Protocol Suite

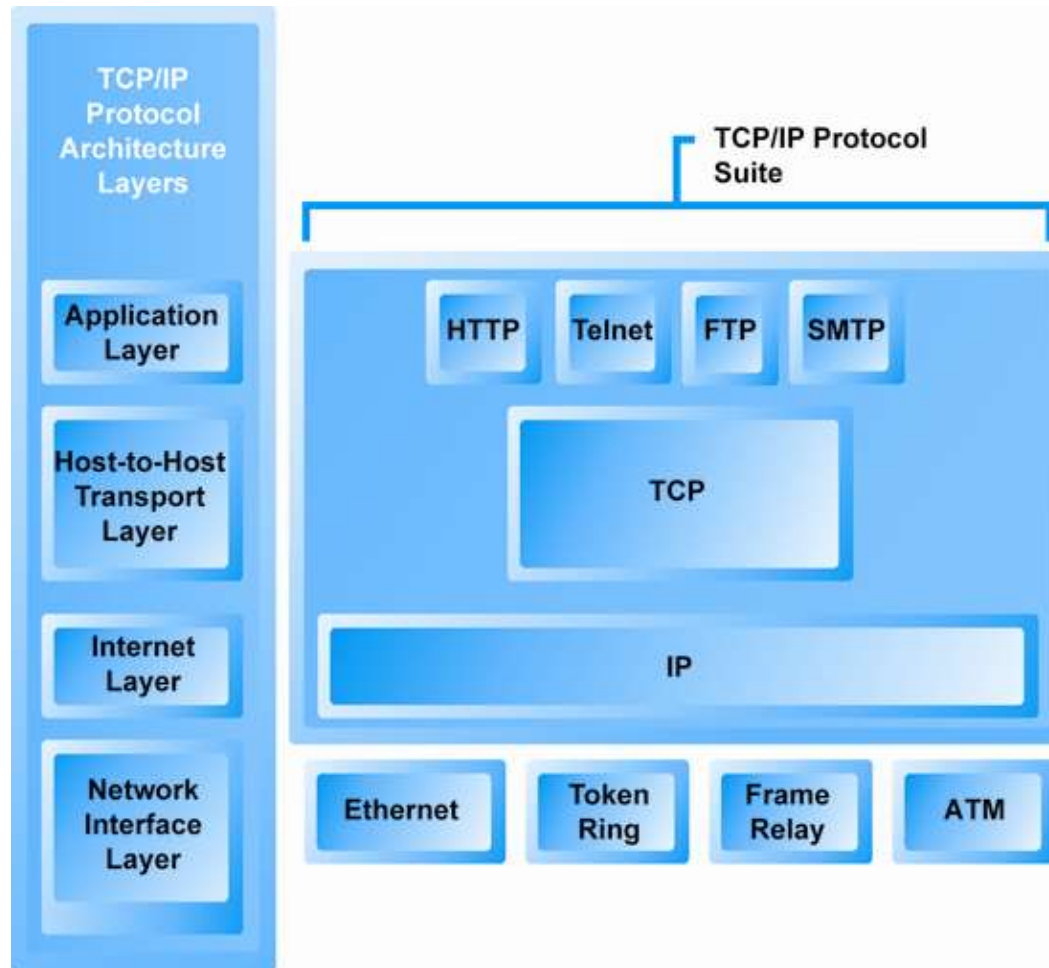


Figure 3.4, Page 119



Internet (IP) Addresses

■ IPv4

- ❖ 32-bit number
- ❖ Four sets of numbers marked off by periods:
201.61.186.227 or 220.216.178.6
 - Class C address: Network identified by first three sets, computer identified by last set

■ IPv6

- ❖ 128-bit addresses, able to handle up to 1 quadrillion addresses (IPv4 can handle only 4 billion)

Routing Internet Messages: TCP/IP and Packet Switching

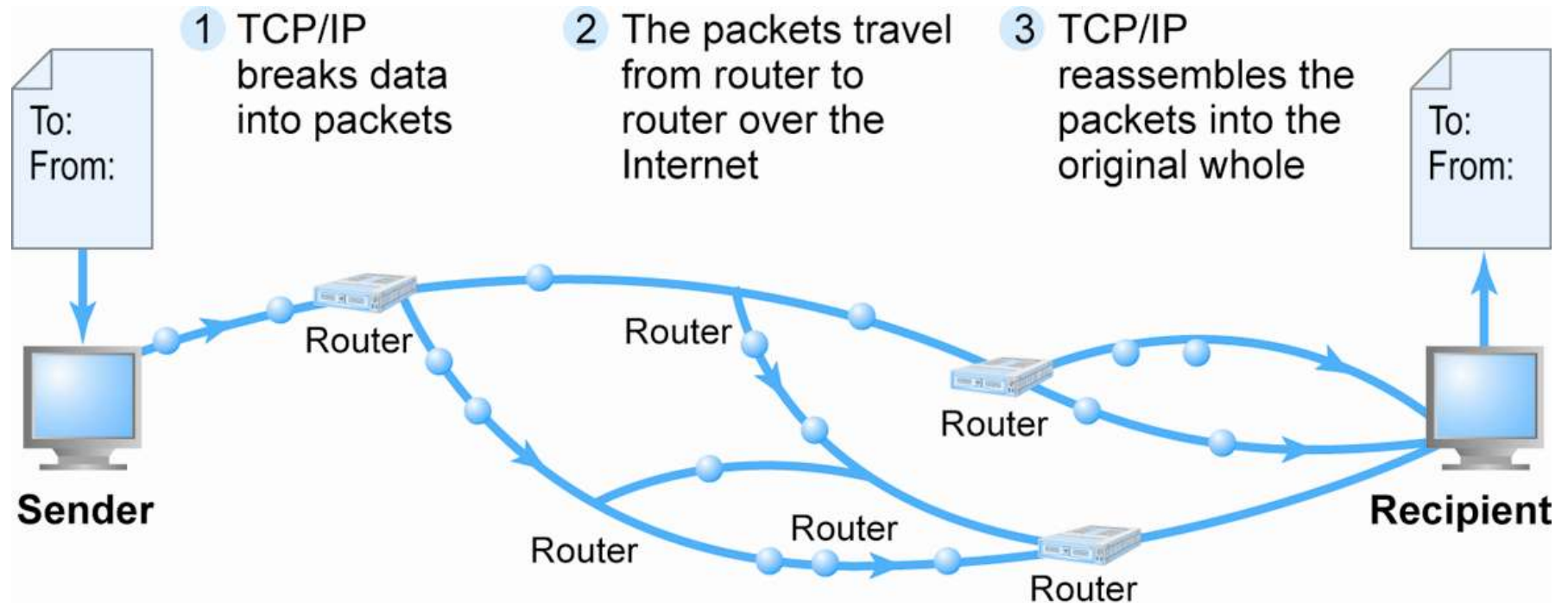


Figure 3.5, Page 120



Domain Names, DNS, and URLs

- **Domain name (.edu, .com, .org etc) (See figure)**
 - ❖ IP address expressed in natural language
- **Domain name system (DNS)**
 - ❖ Allows numeric IP addresses to be expressed in natural language
- **Uniform resource locator (URL)**
 - ❖ Address used by Web browser to identify location of content on the Web
 - ❖ For example: `http://www.azimuth-interactive.com/flash_test`



Client/Server Computing

- **Powerful personal computers (clients) connected in network with one or more servers**
- **Servers perform common functions for the clients**
 - ❖ Storing files
 - ❖ Software applications
 - ❖ Access to printers, and other peripherals



Other Internet Protocols and Utility Programs

■ Internet protocols

- ❖ HTTP
- ❖ E-mail: SMTP (Simple Mail Transfer Protocol), POP3 (Post Office Protocol 3), IMAP (Internet Message Access Protocol) alternative to POP
- ❖ FTP, Telnet, SSL (Secure Sockets Layer)/TLS (Transport Layer Security) newer alternative to SSL

■ Utility programs

- ❖ Ping
- ❖ Tracert is one of several route/path tracing utilities



Internet Service Providers (ISPs)

- **Provide lowest level of service to individuals, small businesses, some institutions**

- **Types of service**

- ❖ **Narrowband (dial-up)**

- ❖ **Broadband**

- Digital Subscriber Line (DSL)

- Cable modem

- T1 and T3

- Satellite



Intranets

■ Intranet

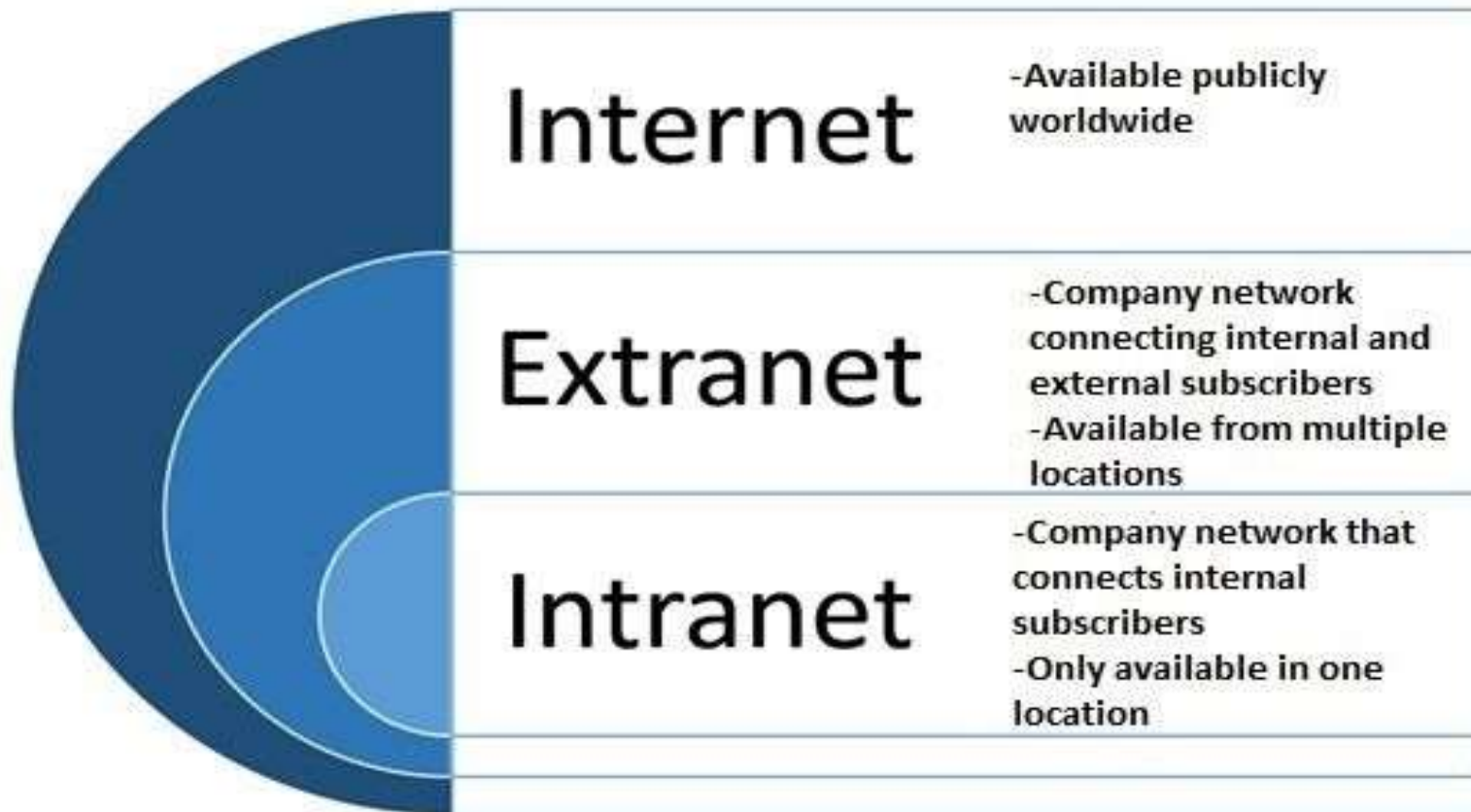
- ❖ TCP/IP network located within a single organization for communications and processing
- ❖ Used by private and government organizations for internal networks
- ❖ All Internet applications can be used in private intranets

Extranet

■ Extranet

- ❖ An extranet is a private network that enterprises use to provide trusted third parties -- such as suppliers, vendors, partners, customers and other businesses -- secure, controlled access to business information or operations.

Internet vs Extranet vs Intranet





Who Governs the Internet?

■ Organizations that influence the Internet and monitor its operations include:

- ❖ Internet Corporation for Assigned Names and Numbers (ICANN)
- ❖ Internet Assigned Numbers Authority (IANA)
- ❖ Internet Engineering Task Force (IETF)
- ❖ Internet Research Task Force (IRTF)
- ❖ Internet Engineering Steering Group (IESG)
- ❖ Internet Architecture Board (IAB)
- ❖ Internet Society (ISOC)
- ❖ Internet Governance Forum (IGF)
- ❖ World Wide Web Consortium (W3C)
- ❖ Internet Network Operators Groups (NOGs)



Limitations of the Current Internet

■ Bandwidth limitations

- ❖ Slow peak-hour service

■ Quality of service limitations

- ❖ Latency – delays due to uneven flow of info packets

■ Network architecture limitations

- ❖ Identical requests are processed individually
(i.e.. Downloading a music track)

■ Wired Internet

- ❖ Copper and expensive fiber-optic cables



Wireless Internet Access Network Technologies

■ Wi-Fi (Wireless Fidelity)

- ❖ High-speed, fixed broadband wireless LAN (WLAN)
- ❖ Wireless access point (“hot spots”)
- ❖ Limited range but inexpensive
- ❖ For-profit Wi-Fi networks: Boingo, AT&T Wi-Fi Services

■ WiMax (Worldwide Interoperability for Microwave Access)

- ❖ High-speed, medium range broadband wireless metropolitan area network (e.g., Baltimore, Portland, Houston, Atlanta)

■ Bluetooth

- ❖ Personal connectivity between devices and to Internet
- ❖ Low-speed, short range connection



Difference between WiFi and WiMAX

Feature	WiFi	WiMAX
Standard	802.11a/b/g/n	802.16d/e
Data rate (MAX)	300 Mbps	70Mbps
Transmission distance (MAX)	300m	50Km
Operating Frequency	2.4 GHz and 5GHz	2-11 GHz
Channel Bandwidth	20 to 25MHz	Ranging from 1.25 to 20 MHz
Encryption	RC4 and Advanced Encryption Standard (AES)	Triple Data Encryption Algorithm (3 DES) and Advanced Encryption Standards (AES)

Wi-Fi Networks

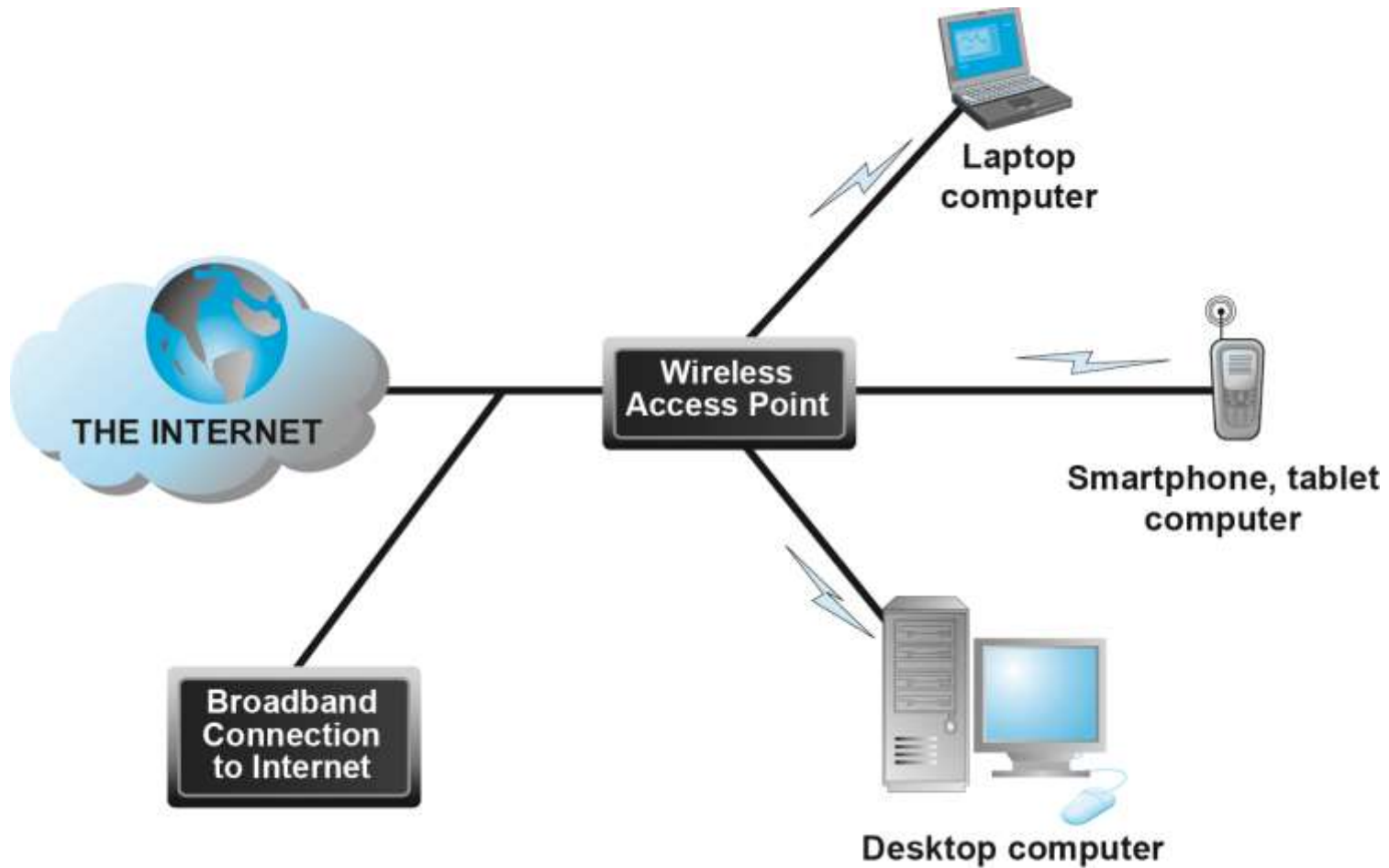


Figure 3.15, Page 145



Hypertext

■ Text formatted with embedded links

- ❖ Links connect documents to one another, and to other objects such as sound, video, or animation files

■ Uses Hypertext Transfer Protocol (HTTP) and URLs to locate resources on the Web

- ❖ Example URL:
<http://megacorp.com/content/features/082602.html>



Markup Languages

■ Hypertext Markup Language (HTML)

- ❖ Fixed set of pre-defined markup “tags” used to format text
- ❖ Controls look and feel of Web pages
- ❖ HTML5 the newest version

■ eXtensible Markup Language (XML)

- ❖ Designed to describe data and information
- ❖ Tags used are defined by user



Web Servers and Web Clients

■ Web server software

- ❖ Enables a computer to deliver Web pages to clients on a network that request this service by sending an HTTP request
- ❖ Apache, Microsoft IIS
- ❖ Basic capabilities: Security services, FTP, search engine capabilities, data capture (e.g. using bar code reader)

■ Web server

- ❖ May refer to either Web server software or physical server
- ❖ Specialized servers: Database servers, ad servers, and so on

■ Web client

- ❖ Any computing device attached to the Internet that is capable of making HTTP requests and displaying HTML pages



Web Browsers

- **Primary purpose to display Web pages**
- **Internet Explorer—54% of market**
- **Mozilla Firefox—20%**
 - ❖ Open source
- **Other browsers**
 - ❖ Google Chrome—19%
 - ❖ Apple's Safari—5%

Search Engines

- **Identify Web pages that match queries based on one or more techniques**
 - ❖ Keyword indexes, page ranking
- **Also serve as:**
 - ❖ Shopping tools
 - ❖ Advertising vehicles (search engine marketing)
 - ❖ Tool within e-commerce sites
- **Outside of e-mail, most commonly used Internet activity**

How Google Works

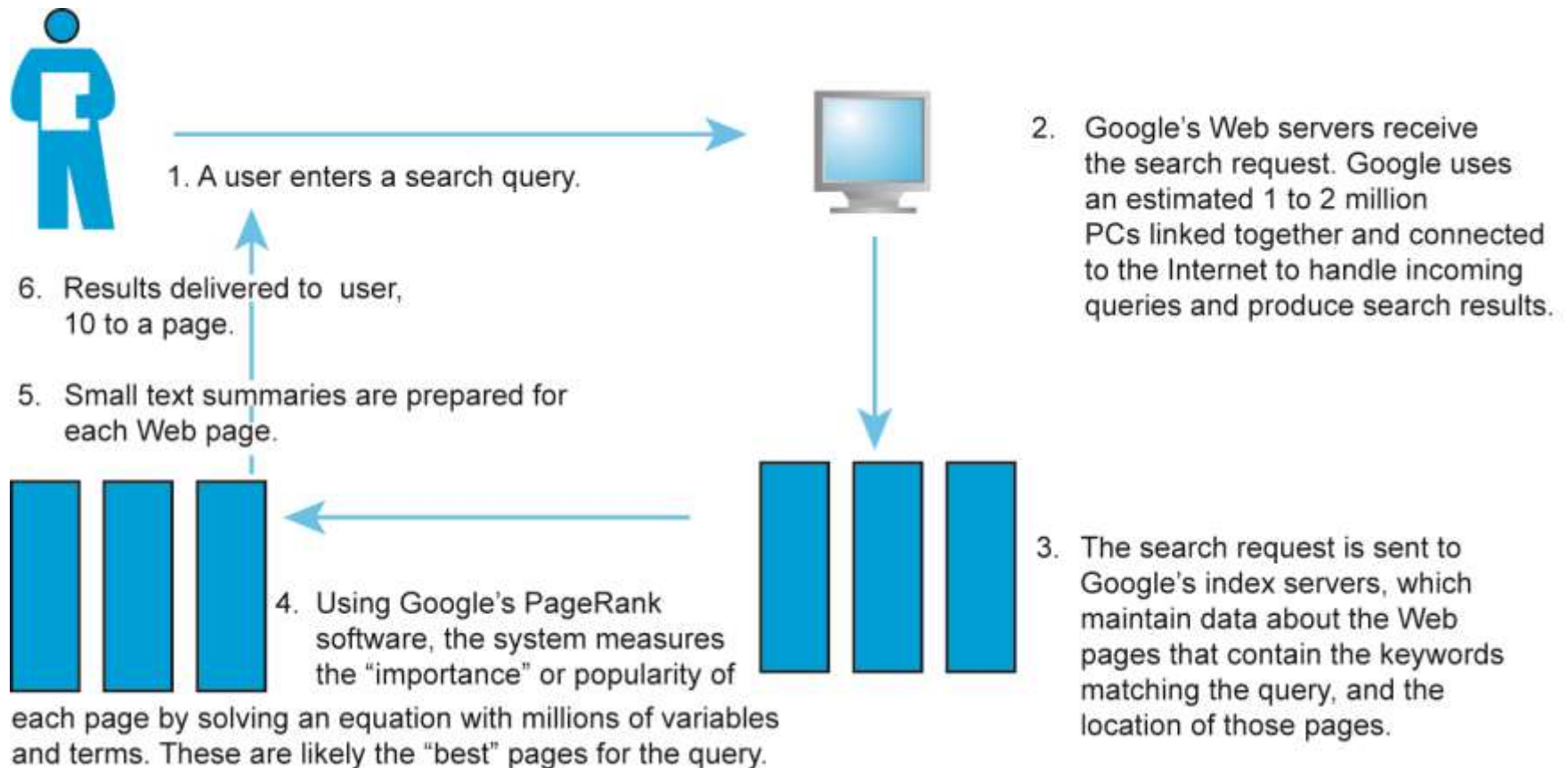


Figure 3.20, Page 161



Online Forums and Chat

■ Online forum

- ❖ Also known as a message board, bulletin board, discussion board, discussion group, board, or forum
- ❖ Web application that enables Internet users to communicate with one another, although not in real time
- ❖ Members visit online forum to check for new posts

■ Online chat

- ❖ Similar to IM, but for multiple users
- ❖ Typically, users log into chat room



Streaming Media

- Enables music, video, and other large files to be sent to users in chunks so that when received and played, file comes through uninterrupted
- Allows users to begin playing media files before file is fully downloaded



Cookies

- **Small text files deposited by Web site on user's computer to store information about user, accessed when user next visits Web site**
- **Can help personalize Web site experience**
- **Can pose privacy threat**



That's All for Today